

Candidate:

Praveena

Date:

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GitHub Profile:

github.com/Praveena32

Primary Stack:

TypeScript, Node.js, React, Prisma

1. Verified Live Deployment & Repository Links

Project Component	Target Live URL	Verified Health
GitHub Repository (Mono-repo)	https://github.com/Praveena32/techloom-assessment	Public & Up-to-date
Task 01: POS Cashier UI	https://techloom-pos-one.vercel.app	HTTP 200 OK
Task 01: POS Backend API	https://techloom-pos-api-delta.vercel.app/health	HTTP 200 OK
Task 02: E-Commerce Storefront UI	https://techloom-store.vercel.app	HTTP 200 OK
Task 02: E-Commerce Backend API	https://techloom-store-api.vercel.app/health	HTTP 200 OK

2. Section 01: POS Order & Inventory System (/task-01)

- Concurrency-Safe Atomic Stock Decrement:** Implemented conditional updates ensuring race conditions are resolved atomically with **zero overselling**.
- 5-Minute Reservation TTL:** Holds items upon checkout entry with automated background sweeper reverting abandoned orders.
- Payment Lifecycle & Idempotency:** Explicit handling for `SUCCESS`, `FAILURE`, and `TIMEOUT` with duplicate checkout protection (`409 Conflict`).
- Automated 50-Request Concurrency Stress Test:**

Total Requests: 50 | Successful Reservations: 5 | Out of Stock Rejections: 45 | Overselling: 0 (Execution T

3. Section 02: E-Commerce Storefront & Payment System (/task-02)

- **Product Discovery:** Debounced real-time keyword search, category filters, dynamic price slider in **LKR**, and in-stock toggle.
- **Product Details & Cart:** Modal view with high-res galleries, technical specifications, and cart management drawer.
- **Pre-Payment Reservation:** Locks stock prior to payment capture to eliminate cart hijacking.
- **Automated Refunds & Inventory Restoral:** "My Orders" customer portal supports 1-click refunds on paid orders, logging refund transactions and returning physical stock to the active catalog immediately.

4. Evaluation Criteria Matrix

Criterion	Engineering Implementation	Result
Concurrency Handling	Atomic conditional decrements (<code>UPDATE ... WHERE stock >= qty</code>)	Verified: 0 Overselling
Reservation TTL	5-minute timed lock with periodic background sweeper (15s interval)	Verified: Auto-restoration
Payment Handling	Three-way outcome simulator (Success, Failure, Timeout) + Idempotency	Verified: 409 on duplicates
Refunds & Cancellations	ACID database transactions restoring stock on cancellation/refund	Verified: 100% Stock recovery
Discovery UX	Faceted search, categories, and price filtering in LKR	Verified: Instant responsive UI

Note on Evaluator Testing: Both live applications are connected to real backend APIs. An interactive Concurrency Test runner is embedded directly in the Task 01 UI for live evaluation.